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Unified Network Forensics Approach for Detecting Malicious Traffic and Data Leakage

PBL Review Report – I

Submitted To

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Problem Statement

Confidential data transmitted over computer networks is vulnerable to interception, exfiltration, and compromise, often escaping detection by conventional monitoring tools such as firewalls and intrusion detection/prevention systems. Attackers increasingly conceal suspicious activities within otherwise normal-looking network traffic, utilizing advanced techniques like encrypted tunnels, protocol obfuscation, or covert data transfers. The challenge lies in proactively identifying abnormal traffic patterns amidst high data volatility and large traffic volumes; such threats frequently exploit unknown or zero-day vulnerabilities, which elude signature-based defenses.^{[1][2][3][4]}

Objectives

- **Capture Live Network Traffic:** Obtain real-time network packet streams to enable precise post-event forensic analysis, focusing particularly on packets that exhibit unusual behavior or characteristics.^[5]
- **Identify Suspicious Transfers:** Detect and flag anomalies in communication patterns, such as unexpected burst uploads, frequent small-sized packets indicative of command-and-control traffic, and outbound traffic to suspicious or unrecognized destinations.
- **Correlate Multi-Trace Evidence:** Integrate packet-level evidence across multiple network segments and timeframes to reconstruct a comprehensive attack narrative, essential for both incident response and legal investigation.^{[3][7]}
- **Summarize Forensic Findings:** Present analytical results in formats suitable for technical and non-technical stakeholders, with an emphasis on clear documentation and forensic chain of custody standards.^[3]

Engineering Systems and Tools

Network forensics investigations leverage a layered suite of tools led by Wireshark, an open-source packet analyzer capable of deep inspection of over 850 network protocols.^{[1][5]}

Tool/Feature	Function	Forensic Role	Skill Level Required
Wireshark (GUI)	Interactive, granular packet analysis	Core investigation platform	Intermediate to Advanced
TShark (CLI)	Command-line analysis	Automated/scripted batch analysis	Advanced
Dumpcap	High-speed capture engine	Background traffic acquisition	Basic to Intermediate
Editcap/Merg ecap	Capture file editing/merging	Removal of extraneous data, file prep	Intermediate
Filters	Dynamic packet filtering	Focuses on packets matching key indicators (e.g. suspected exfiltration, C2)	Intermediate
Expert Info	Protocol violation detection	Identifies malformed/abnormal traffic	Intermediate

Supporting systems include network taps, SPAN ports, forensic workstations, compliant evidence storage, and reporting/documentation systems. Modern forensics environments should facilitate integration with SIEM platforms and external threat intelligence, emphasizing evidence integrity and legal chain-of-custody.^{[5][3]}

Variables and Parameters

- Packet Features: Byte size, transport protocol type, timing intervals, TCP/UDP flags, application protocol markers.^[5]
- Transfer Attributes: Volume, frequency, directional flow analysis (e.g. sudden outbound spikes), session duration and recurrence.^[2]
- Source/Destination Metrics: IP addresses, autonomous system/geographic location, known blacklisted endpoints.^[4]
- Parameter Thresholds: Excessive data transfer rates, anomalous port usage, off-hours communication patterns, and protocol compliance deviations.

Existing Processes and Solution Methods

Traditional Monitoring (Firewall/IDS/IPS): Rule-based and signature-driven defenses excel at identifying known attacks but generally fail against new threat vectors and polymorphic malware.^{[4][1]}

Signature-Based Detection: Streamlined for established patterns; offers low false positives but performs poorly with novel or targeted attacks.^{[2][1]}

Anomaly-Based Detection: Utilizes behavioral baselines and statistical anomaly detection models for surfacing unusual network activity; more effective against zero-day threats but requires continual tuning to reduce false positives.

Manual Packet Inspection (Wireshark): Provides unparalleled detail through interactive analysis, enabling identification of subtle indicators, suspicious payloads, and covert communication methods. However, manual inspection is resource-intensive and impractical for large scale environments unless augmented by automation.^[5]

SIEM & Log Correlation: Aggregates network telemetry and logs for contextual analysis of security events, supports evidence correlation, and incident response workflows but may miss packet-level evidence crucial for forensic depth.^[3]

Data Loss Prevention (DLP): Employs flow tracking and content analysis, effective for policy violation monitoring but may be circumvented by encryption or tunneling.

Compare and contrast alternative solution processes to select the best process

Solution Method	Detection Type	Real-time Capability	False Positives	Scalability	Evidence Quality	Unknown Threats	Resource Requirement	Legal Suitability
Traditional Monitoring	Known threats	Yes	Medium	High	Low	Poor	Medium	Limited
Signature-based Detection	Known patterns	Yes	Low	High	Limited	Poor	High	Limited
Anomaly-based Detection	Behavioral	Yes	High	Medium	Strong	Good	Medium	Medium
Manual (Wireshark)	Deep detailed	Post capture	Very Low	Low	Excellent	Strong	Skilled analysts	Strong
SIEM/Log Correlation	Multi-source	Yes	Medium	High	Good	Good	High	Medium
Unified Approach (Wireshark+ Automation)	All threats	Yes post capture	Low	Medium	Excellent	Strong	High	Strong

Given the trade-offs, a unified forensic methodology combining Wireshark's rich packet-level analysis and automated anomaly/behavioral detection offers optimal coverage for both unknown and known threats, balancing evidence integrity, scalability, and actionable detection. Manual approaches remain the gold standard for legal evidence strength but are best reserved for high-value/critical incident investigation and evidence review.^{[2][3][5]}

Technical and Non-Technical Information

Technical Domains:

- Deep packet inspection (DPI) for header/payload anomalies.^[5]
- Multi-protocol support including DNS, HTTP(S), FTP, SMTP, encrypted channels.^[3]
- Pattern recognition through statistical distributions, frequency analysis, and feature correlation using time-based and session-based metrics.^[4]

Non-Technical Domains:

- Legal Admissibility: Chain of custody, documentation, integrity verification, tool validation standards, compliance with privacy regulations.^[3]
- Organizational Policy and Risk: Integration with incident response, stakeholder reporting, business risk translation, and staff training for ongoing forensic readiness.^[3]

Bridging both domains enables evidence to move seamlessly from technical discovery to effective legal presentation and risk management.

Methodology Framework Implementation (OSCAR Model)

1. Obtain Information: Comprehensive acquisition of incident, network, and organizational context; documentation of incident timeline and network architecture.^[3]
2. Strategize: Define forensic scope, resource prioritization, investigation team roles, and communication protocols, aligned with organizational and legal requirements.^[3]

3. Collect Evidence: Packet capture (live and historical) using Wireshark and related utilities, log and artifact collection, strict chain-of-custody maintenance.^{[5][3]}
4. Analyze: Use Wireshark's analytical suite to review packet flows, reconstruct communications, correlate multiple traces, and identify malicious or anomalous behavior through advanced filtering and session reconstruction.^[3]
5. Report: Produce concise, well-documented reports detailing evidence, methodology, findings, non-technical summaries for executives, and technical annex for IT/security/legal teams. Ensure reproducibility and evidentiary standards for court acceptance.^[3]

Advanced Detection Techniques and Wireshark Filters

- Behavioral Filters: `tcp.flags.syn==1 && tcp.flags.ack==0` (port scanning), `dns.qry.name contains suspiciousdomain`, `http.request.uri matches parameter=value$`.^{[5][6]}
- Exfiltration Indication: `frame.len > 1500 && tcp.len > 1460` (large file transfers), detecting non-standard protocol usage, monitoring beginnings of SSL/TLS handshakes to highlight encrypted tunnel establishment.^[6]
- Network Anomalies: Failed connections (`tcp.flags.reset==1`), time-based triggers for after-hours communications, geo-IP filters for abnormal source/destination behavior.

Legal and Compliance Considerations

Forensic investigations must comply with technical and non-technical legal requirements:

- Technical: Maintain strict chain of custody, employ validated tools, ensure cryptographic data integrity (hashes of capture files), and guarantee analysis reproducibility.^[3]

- Legal: Possess appropriate authorization for monitoring/data collection, respect privacy regulations, prepare evidence and expert testimony for court, maintain exhaustively documented procedures.^[3]

Conclusion and Implementation Recommendations

Network forensics, when unified through Wireshark and robust methodology frameworks, provides comprehensive, reliable detection of malicious traffic and data exfiltration. Success demands ongoing technical skill development, tool enhancement, business and legal integration, and continuous adaptation to emerging threats. Proper methodology, such as OSCAR, combined with Wireshark-based packet analysis, forms a resilient foundation for both cyber risk mitigation and legal investigation.^[3]

References

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